The simulator platform

Before describing the users, it is necessary to describe the tool they will interact with, because the mental model is always relative to the product.

The Bayer Yield Simulator is a three-page web platform that processes agronomic inputs against historical FieldView data filtered to four Brazilian states and two crops. Each page has a distinct purpose and users move through them in sequence.

Page	Name	Purpose
Page 1	**Home**	Explains what the simulator does, which agronomic questions it answers, and how it should be used. Provides scope information (states and crops covered) and guides the user toward Page 2.
Page 2	**Variable Configuration**	Dedicated to the user reviewing and adjusting the available variables. Pre-filled from FieldView historical data where available. The three inputs that must be confirmed before the simulation runs are: state (geographic scope), crop, and planting date. All other variables are pre-filled and optionally overridable. Triggers the simulation when the user is satisfied with the configuration.
Page 3	**Behavior Demonstration**	Dedicated to presenting the simulation output: the three productivity scenarios, calculated metrics, climate context retrieved from FieldView, the dataset similarity search results, and the explanatory agronomic recommendation. Supports re-running the simulation by returning to Page 2.

On Page 3, the simulator returns a structured output containing: three productivity scenarios (pessimistic, most likely, optimistic); a calculated Coefficient of Variation of Longitudinal Seed Distribution; a Phenotypic Plasticity indicator; climate context data automatically retrieved from the FieldView database for the selected field and planting window; a dataset similarity search that identifies historical records in the FieldView database with conditions comparable to the user's current configuration; and a plain-language agronomic recommendation text.

Output element (Page 3)	What the user sees
Three productivity scenarios	Pessimistic, Most Likely, and Optimistic, each with a yield estimate in sc/ha and a confidence band.

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CV of Seed Distribution	A percentage expressing the uniformity of seed spacing; derived from seed_actual_population_per_hectare and field conditions.
Phenotypic Plasticity	An indicator of the crop capacity to compensate for sub-optimal conditions; depends on hybrid, soil order, and planting window.
Climate context	Temperature, rainfall, and wind speed data retrieved automatically from the FieldView historical database for the field's location and planting window. The user sees this data for context; it is not entered manually.
Dataset similarity search	A query over the FieldView historical dataset identifying records whose state, crop, soil profile, planting window, and seed population closely match the user's current configuration. Results are presented as comparable cases with their observed outcomes, giving the user empirical grounding for the simulated scenarios.
Narrative recommendation	A 3 to 5 sentence plain-language text telling the farmer what approach to follow given the simulated scenario and the comparable cases found.
Immutable premises	Fixed agronomic thresholds embedded in the model (e.g. minimum viable pH, reference population ranges), displayed but not editable.
Auditable metrics	Values the user can inspect and adjust to understand how the model reached its output (e.g. planting date, soil drainage class).

Primary user: the crop manager

Profile

Name	João (representative persona)
Role	Crop manager or senior agronomist on a medium-to-large farm
States in scope	Minas Gerais, Paraná, Mato Grosso do Sul, or Goiás
Crops	Corn (summer or safrinha) and/or soybeans
Farm scale	500 to 5,000 ha under management; typically 3 to 8 fields active per season
FieldView usage	Active user; familiar with the planting records and soil profile data
Device	Primarily desktop or laptop in the office; occasionally tablet in the field

Technical level	Agronomically trained; comfortable with percentages and yield units; not a data scientist
Decision timing	Pre-season planning (Oct to Nov for soybeans; Jan to Feb for safrinha corn)

Goals when opening the simulator

- Get a realistic yield range for the season before committing to a hybrid or planting date.
- Understand whether the planned seed population is well matched to the field's soil conditions.
- Have a defensible answer ready when the farm owner asks how this season is looking.
- Identify the one or two variables most worth improving to move from the pessimistic to the most likely scenario.

Frustrations with the current situation

- Yield estimates today come from memory, neighbour conversations, or regional bulletins, none specific to this field's soil profile and this year's planting date.
- The FieldView platform shows data but does not tell him what to do with it.
- He has to open multiple separate tabs and mentally combine them, a process that takes 40 to 90 minutes and produces a number he is not confident in.

Mental model: what João believes when he opens the simulator

Understanding this mental model is essential because the interface will either confirm or contradict it. When it confirms, the user flows through with no friction. When it contradicts, the user stops, questions the tool, and often abandons it.

Belief 1: More inputs equal a better answer

João believes that a more detailed configuration will produce a more reliable output. He is accustomed to filling in detailed forms in FieldView and agricultural management systems. He will not be satisfied by a two-field form; he expects to see state, crop, planting date, soil order, pH range, drainage class, and seed product all present and labeled, even when most values are already pre-filled.

Page 2 implication: the variable configuration page must feel comprehensive. Defaulting to average values without letting him review and override them will feel like the tool is guessing.

Each input field must be visible and labeled. The three hard-blocker inputs (state, crop, planting date) must be visually prominent. All remaining fields are pre-filled from FieldView data and optionally editable.

Belief 2: A single number is not enough

João has seen too many seasons deviate from a predicted yield to trust a point estimate. He already thinks in ranges: a good year, a bad year, a middle year. Three scenarios match his cognitive model exactly. He does not need to be taught this framing; he already uses it.

Page 3 implication: the three-scenario output must be the most visually prominent element of the behavior demonstration page. A single headline number followed by a footnote about scenarios will be read as a point estimate. The design must present the three simultaneously, at equal visual weight.

Belief 3: The model should explain itself

João distrusts black boxes. He has encountered crop-management applications that gave him a recommendation with no explanation and felt manipulated. He will not follow a recommendation he cannot trace. He wants to see which variables drove the most likely scenario and verify that they match his field knowledge.

Page 3 implication: every metric in the output section must have a visible label, a numeric value, and a short contextual annotation. The comparable cases from the dataset similarity search reinforce this by providing real historical records that João can validate against his own field experience. The narrative text must reference the specific inputs that drove the recommendation.

Belief 4: The recommendation must translate into a field action

João does not open the simulator to learn agronomy. He already knows agronomy. He opens it to get a decision ratified or challenged. The output he values most is not the yield number; it is the answer to whether he should change anything before he plants. The narrative text is therefore the most important output element from his perspective, even though it is not a number.

Page 3 implication: the narrative recommendation text must appear alongside the three scenarios, not buried below them. It must be written in the second person and must be specific enough that João can act on it without further interpretation.

Belief 5: Fixed parameters should be honest, not hidden

João is a professional. He is not offended by the existence of model assumptions. What offends him is discovering them after the fact. He expects a tool that uses Embrapa reference thresholds or Bayer hybrid response curves to say so, visibly, even if those values cannot be changed.

Page 3 implication: immutable premises must be displayed in the behavior demonstration page, in a clearly labeled section called Model Assumptions, not hidden in a tooltip or an external manual. Auditable metrics must be visually differentiated from fixed premises using distinct iconography.

Belief 6: Comparable cases from the past are more convincing than a model alone

João is empirically minded. He trusts what happened in real fields more than what a model projects. When the simulator shows that fields with a similar soil profile, planting window, and seed population in the same state achieved a certain yield range in previous seasons, he finds this more persuasive than a scenario derived purely from agronomic formulas.

Page 3 implication: the dataset similarity search results must be presented as a distinct section of the behavior demonstration page, clearly separated from the model-generated scenarios. Each comparable case must show the key matching variables and its observed outcome. The section must be explicitly identified as historical records, not projections.

Belief 7: Climate data should be available without burdening the user

João knows that climate conditions are critical to yield outcomes. However, he does not want to manually look up and enter weather data before he can run a simulation. He expects the tool to bring climate context from the same data infrastructure he already uses, without requiring him to fetch it from a separate source. Seeing climate data appear automatically confirms that the system knows his field, which reinforces his trust in the output.

Page 3 implication: climate context (temperature, rainfall, wind speed for the relevant planting window) must be displayed on Page 3 as automatically retrieved data, clearly labeled as 'Retrieved from FieldView historical data'. The user must not be asked to enter or confirm any climate variable on Page 2. The presence of this data on Page 3 reinforces João's trust in the model without adding friction to his configuration workflow.
